

Gradable Answers and Inquisitive Progress

With an Application to Inquisitive Norms

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Claim 1

Questions being answered share some structural similarities to gradable adjectives.

Claim 2

Similarity suggests a particular semantics for questions. Consistent with established accounts.

Claim 3

Adopting this semantics has direct effects on inquisitive norms and suggests a notion of inquisitive progress.

- Philosophical speech act content analysis makes use of a question under discussion (e.g. Bowker, 2022; Friedrich, 2025; Hoek, 2018; Schoubye and Stokke, 2016).
- Formal semanticists want to treat declarative and interrogative phrases in a unified framework (e.g. Ciardelli et al., 2018; Groenendijk and Roelofsen, 2009).
- Inquiry is often characterized as having a question-directed attitude (e.g. Friedman, 2013; Habgood-Coote, 2022; Whitcomb, 2017).
- Epistemic theories make use of questions to analyze the content of propositional attitudes (e.g. Beddor and Goldstein, 2023; Hoek, 2022; Leitgeb, 2014; Yalcin, 2018).
- Questions are structurally related to subject matter (e.g. Fine, 2020; Yablo, 2018).

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Observation

Clearly, some answers are better than others.

- “It’s the 24th of May, 2026” is a better answer than “It’s the 24th” to “What date is it?”
- “Only Peter, Paul, and Mary” is a better answer than “Mary, don’t know about the others” to “Who’s coming to dinner?”

Suggests: more informative answers are better. (Can you think of another way in which one answer is better than a second?)

Question

Could 'good answer' be semantically similar to gradable adjectives?

- Examples: tall, flat, bent, etc. (Hawthorne & Logins, 2021)
- Admissibility of comparative constructions
 - 'x is flatter than y', 'x is more regrettable than y', 'x is as regrettable as y' etc.
- Felicity of 'how F' questions
 - 'How tall is she?', 'How flat is it?', etc.
- Admissibility of degree modifiers
 - 'x isn't flat at all', 'x is extremely regrettable', 'x is slightly wet', 'x is somewhat bent', 'y is completely confident', etc.

Question

Could 'good answer' be semantically similar to gradable adjectives?

- Admissibility of comparative constructions
 - "It's the 24th of May, 2026" is a better answer than "It's the 24th"
- Felicity of 'how F' questions.
 - How good is that answer? How well does that answer your question?
- Admissibility of degree modifiers.
 - That's a very good answer! (But compare: That only partially answers the question, that's a completely good answer)

Observation

Yet often, we can't tell which answer is better.

- “There were at least two” is neither a better or worse answer than “One of them was male” to “Who committed the crime?”
- “Peter, don't know about the others”
is neither a better or worse answer than
“Mary, don't know about the others” to “Who's coming to dinner?”

Suggests: incommensurability w.r.t. informative content

Cavete:

- 'good answer' vs. 'is answered'
- Isn't 'good anything' trivially gradable, see Kamp, 1975?
- Not the only way in which one answer is better than another.

Standard semantics for graded adjectives requires **a scale** and **a threshold** (Kennedy & McNally, 2005; Lassiter, 2017).

Two questions to answer:

- Where do scale and threshold come from?
- Why incommensurability intra- and inter-question?

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- A question denotes a set of propositions called *q-alternatives*.
- Basic semantic answers: substitute interrogative pronoun by an expression of the right semantic type.
- “Who has committed the crime?”
- “_ has committed the crime.”

(1) Who has committed the crime?

$$q\text{-alt} = \{\text{Alice has committed the crime,} \\ \text{Bob has committed the crime,} \\ \text{Charlie has committed the crime}\}.$$

Observation.

- Such a set of propositions uniquely partitions logical space.

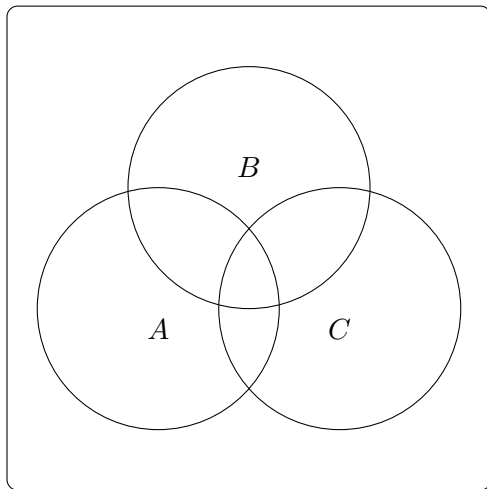


Figure: Who has committed the crime?

Definition

A *complete answer* is a single partition cell.

	ABC	
$AB\bar{C}$	$A\bar{B}C$	$\bar{A}BC$
$A\bar{B}\bar{C}$	$\bar{A}B\bar{C}$	$\bar{A}\bar{B}C$
	$\bar{A}\bar{B}\bar{C}$	

Figure: Alice, but neither Bob nor Charlie, has committed the crime.

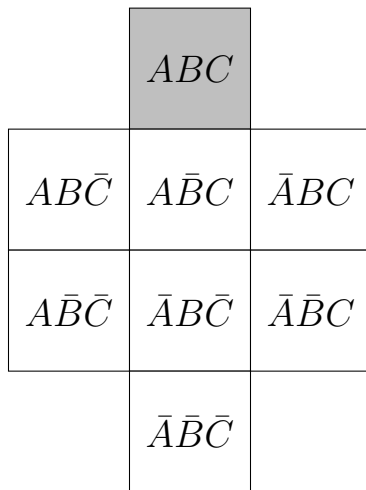


Figure: Alice and Bob and Charlie have committed the crime.

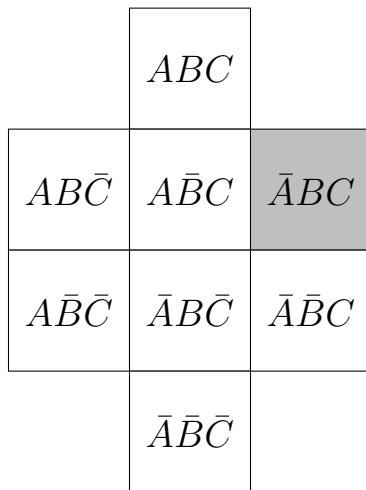


Figure: Bob and Charlie, but not Alice, have committed the crime.

Definition

A *partial answer* is a (non-empty, non-total) union of partition cells

Observations.

- Any complete answer is also a partial answer.
- Partial answers do not, generally, entail the truth or falsity of a q-alternative.
- Partial answers are described by Ciardelli et al. (2018) and Groenendijk and Stokhof (1984, 1997).
- Partial answers as defined here slightly different to Roberts (2012).

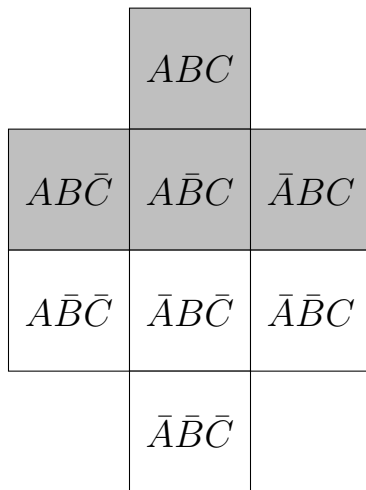


Figure: It was at least two of them! [Union of partition cells shaded gray]

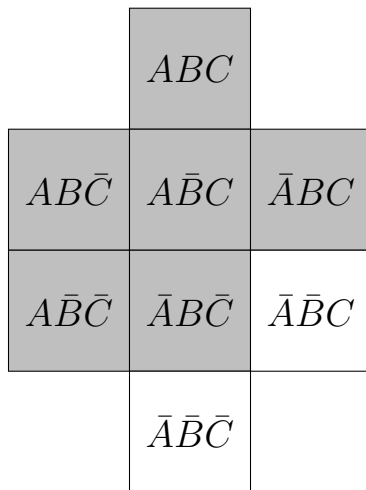


Figure: Alice or Bob committed the crime. [Not deciding whether Charlie was involved, or whether Alice and Bob did it together or one of them separately]

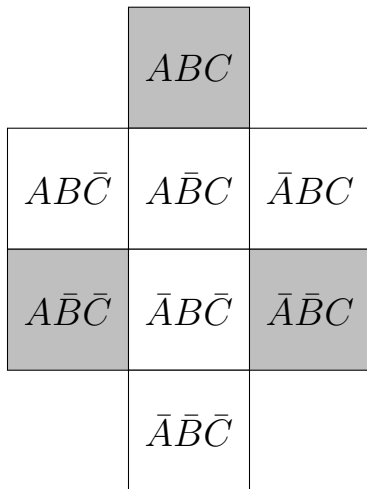


Figure: Alice and Bob and Charlie all committed the crime, or only Alice did, or only Charlie did.

Consider only:

(2) Who has committed the crime?

$q\text{-alt} = \{\text{Alice has committed the crime,}$
 $\text{Bob has committed the crime}\}.$

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(a) A and B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(b) A and not-B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(c) not-A and B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(d) not-A and not-B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(a) A

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(b) not-A

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(c) B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(d) not-B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(a) A iff B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(b) A iff not-B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(a) A or B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(b) A or not-B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(c) not-A or B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(d) not-A or not-B

AB	$A\bar{B}$
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(a) A and B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(b) A

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(c) A or B

Figure: (a) $\Rightarrow$ (b) $\Rightarrow$ (c)

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(a) not-A and not-B

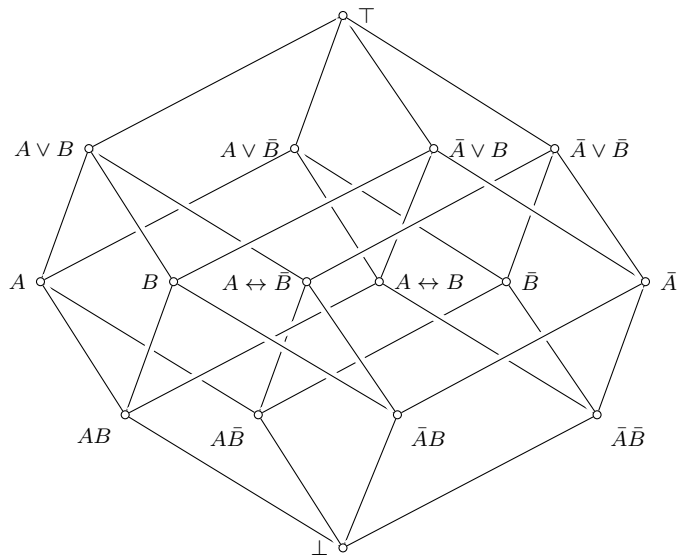
AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

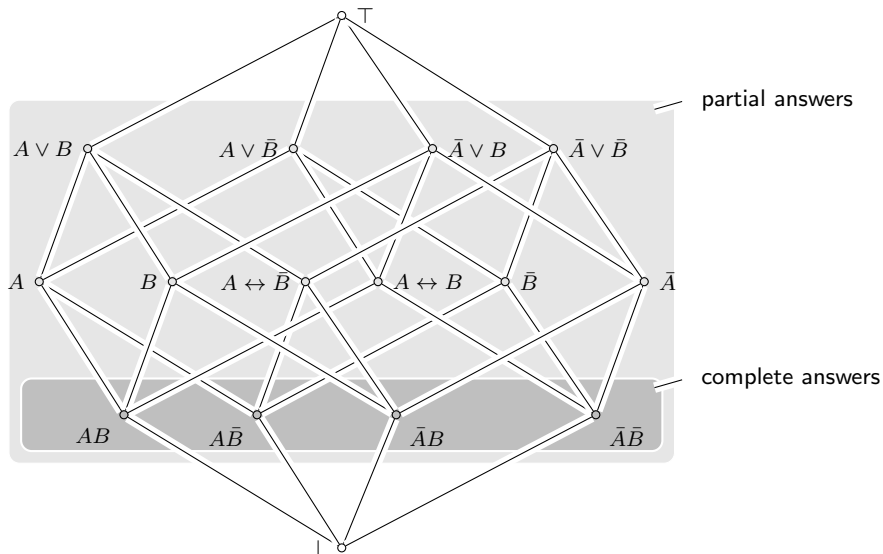
(b) A iff B

AB	$A\bar{B}$
$\bar{A}B$	$\bar{A}\bar{B}$

(c) A or not-B

Figure: (a) $\Rightarrow$ (b) $\Rightarrow$ (c)





Standard semantics for graded adjectives requires a **scale** and a **threshold**.

- **Scale:** partial order by entailment.
 - Explains incommensurability intra- and inter-question.
- **Threshold:** a **set** (of answers).

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- If the goal of inquiry is to settle a question, then coming to believe a more informative answer is progress towards settling the question.
- And thus is inquisitive progress.
- One way of *making progress* in the inquiry.

Detective Holmes is investigating. Who committed the crime? The only suspects are Alice, Bob, and Charlie. Watson exclaims:

- *It was at least two of them!*

Holmes seems to make inquisitive progress coming to believe:

- that it was **exactly** two of them.
- that Alice committed the crime. [Not deciding whether Bob or Charlie did, too]

However, these don't seem to constitute inquisitive progress:

- that it was at least one of them. [No new information]
- that it was at most one of them. [Incompatible information–Revision?]

I make inquisitive progress if I come to believe a more informative answer than before.

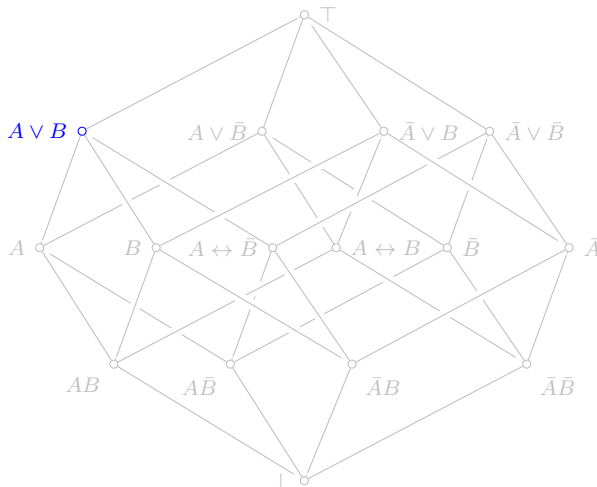


Figure: Suppose I believe that A or B . Coming to believe which answers implies inquisitive progress?

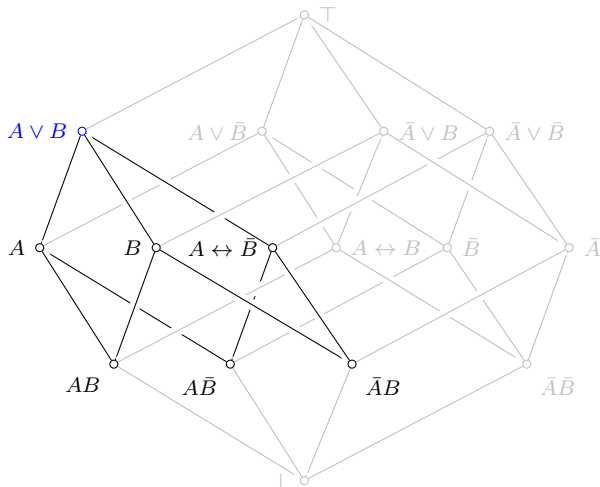


Figure: Suppose I believe that A or B . Coming to believe which answers implies inquisitive progress?

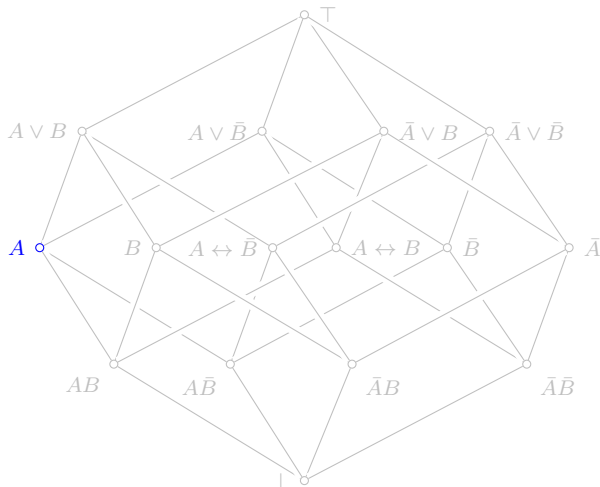


Figure: Suppose I believe that A . Coming to believe which answers implies inquisitive progress?

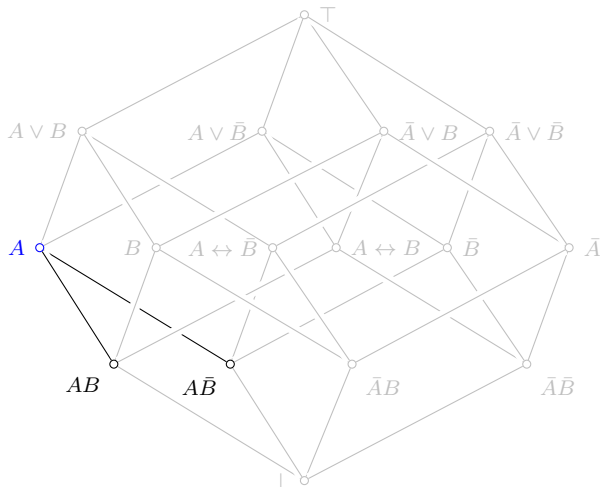


Figure: Suppose I believe that A . Coming to believe which answers constitutes inquisitive progress?

Inquisitive Progress (sufficient condition)

If S believes answer a to q and comes to believe answer b to q , with

- $b \Rightarrow a$,

then S made *inquisitive progress* w.r.t. q .

DBI

One ought not: believe a complete answer to q and inquire into q .

- (3) * I believe it's raining, but I wonder whether it's raining.
- (4) * I believe it's raining, but is it raining?
- Doesn't tell us anything about incomplete answers.
- Desideratum: Generalize DBI to incomplete answers, with the same result for complete answers.¹

¹Sidelining discussion about the plausibility of DBI.

- If I believe incomplete answer a to a question q , may I still inquire into q ?
- It seems so, but only on the supposition that a .

Proposal

Believing an incomplete answer to q does not prohibit inquiry into q , but instead *systematically constrains* it.

Where to go from here?

- Often, we merely find evidence in favor of one answer over the other, without entailing any answer as defined. Probabilistic account called for.
- There are more ways to make inquisitive progress. For example,
 - Refining the question can constitute progress.
 - Revising one's answer in light of new evidence can constitute progress.
 - Merely evidential answers can constitute progress.
- Relate graded notion of answer to *resolving an issue* in inquisitive semantics.

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